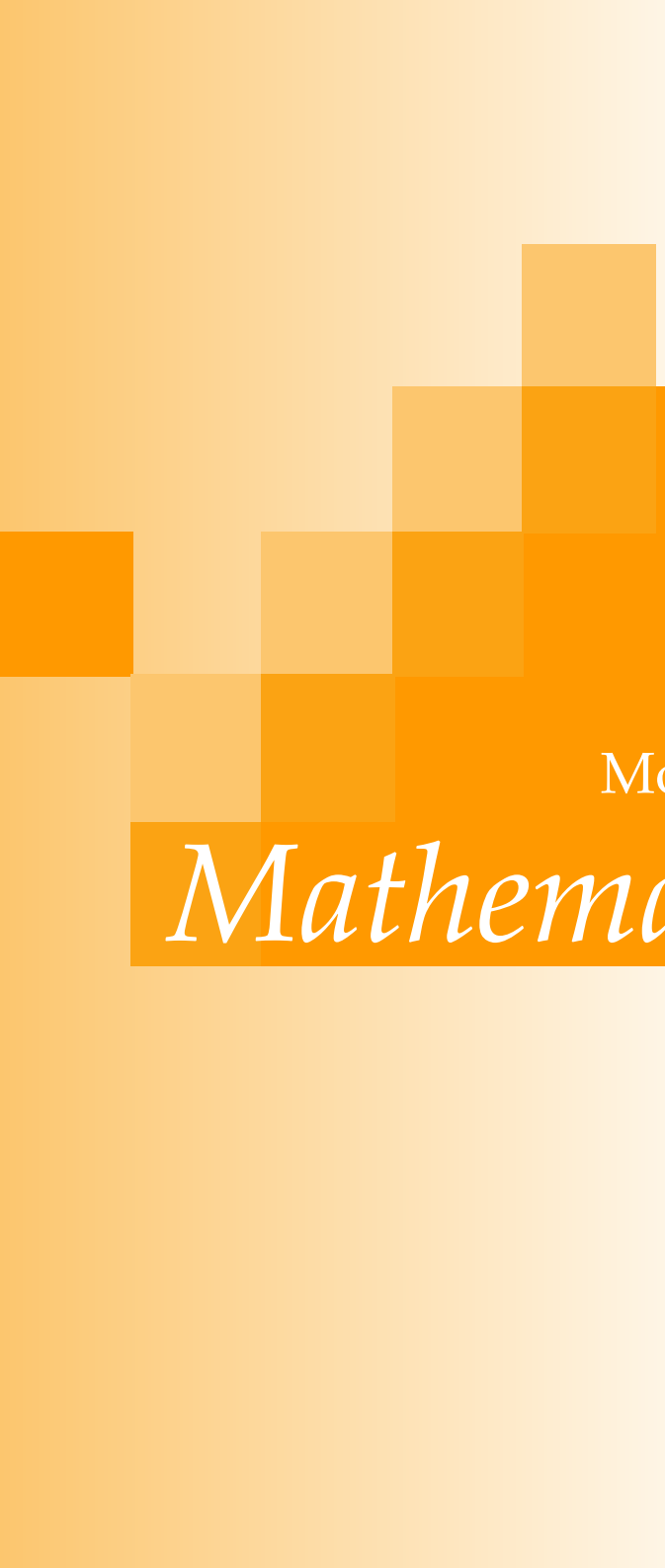




Machine Learning



Module 2

Mathematical Foundation Of ML

System of Linear Equations

- A **system of linear equations** (or **linear system**) is a collection of two or more linear equations involving the same set of variables.
- *an essential field of mathematics, which defines the study of vectors, matrices, planes, mapping, and lines required for linear transformation.*
- the important and primary thing to process the applications of Machine Learning.
- It is also a prerequisite to start learning Machine Learning and data science.
- *it enables ML algorithms to run on a huge number of datasets.*

Representation of linear equations in matrix and vector forms:

Lets take a system of three linear equations in three variables x, y, and z

$$a_1x + b_1y + c_1z = d_1$$

$$a_2x + b_2y + c_2z = d_2$$

$$a_3x + b_3y + c_3z = d_3$$

Matrix Representation

The equations can be written as matrix multiplication form -

$$\overbrace{\begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}}^A \overbrace{\begin{bmatrix} x \\ y \\ z \end{bmatrix}}^X = \overbrace{\begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}}^D$$

or,

$$AX = D$$

Matrix A is called the **coefficient matrix**. If we append the columns of matrix D with matrix A like below the resultant matrix is called **augmented matrix**, denoted by $[A|D]$

$$[A|D] = \begin{bmatrix} a_1 & b_1 & c_1 & d_1 \\ a_2 & b_2 & c_2 & d_2 \\ a_3 & b_3 & c_3 & d_3 \end{bmatrix}$$

Vector Representation

The above linear equations can also be represented in vector form as :

$$x \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} + y \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} + z \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$$

or,

$$ax + by + cz = D$$

where $a = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$, $b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$, $c = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}$, $D = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$ are the column vectors

$$AX = ax + by + cz = D$$

or,

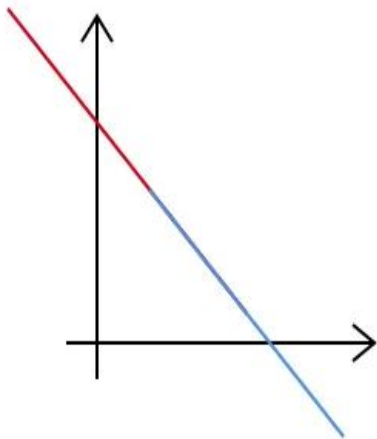
$$\begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = x \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} + y \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} + z \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}$$

Solution set of linear equations

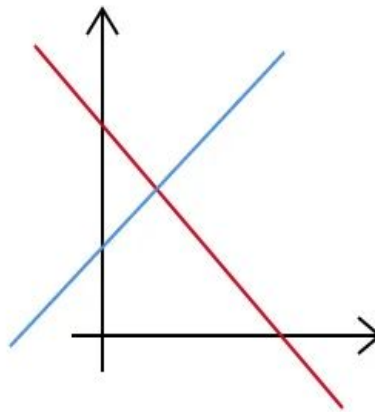
- Every linear system may have only one of three possible number of solutions:
 - 1. The system has a single unique solution.***
 - 2. The system has infinitely many solutions.***
 - 3. The system has no solution.***

Solution set of linear equations

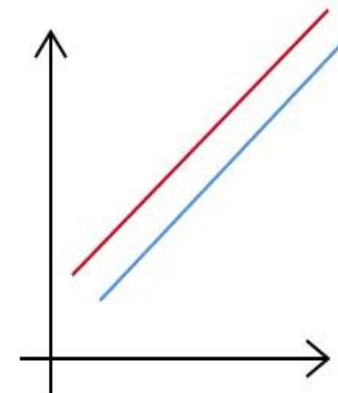
- For a system of two variables (x and y), each linear equation determines a line on the xy -plane.
- The solution set is the intersection of these lines, and is hence either a line, a single point or don't have any common point.



Infinite no of Solution : Line



Unique Solution : point



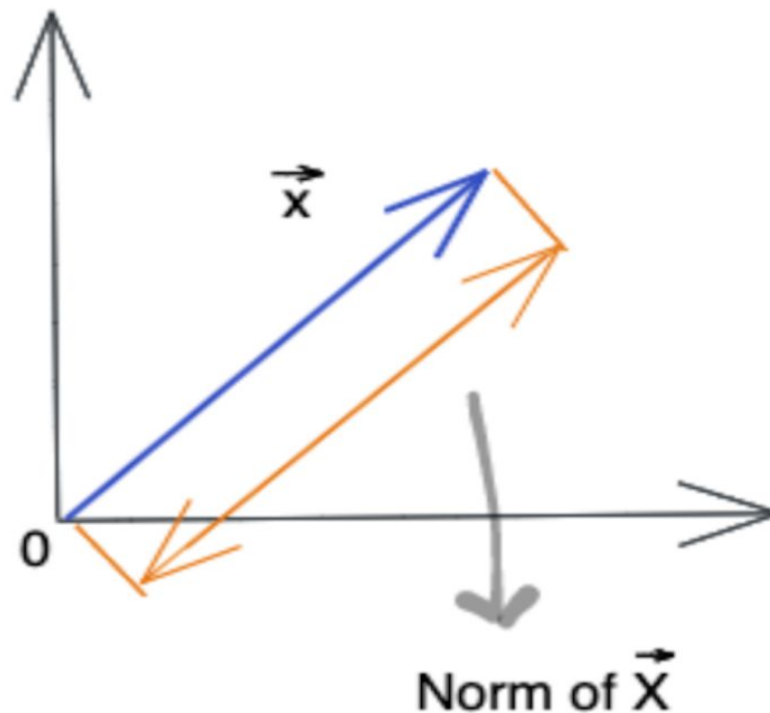
No Solution

Norms

- **What Is a Norm?**

A norm is a way to measure the size of a vector, a matrix.

- In other words, norms are a class of functions that enable us to quantify the magnitude of a vector.
- For instance, the norm of a vector X drawn below is a measure of its length from origin.



Norms

- The length or the magnitude of the vector is known as vector norm or vector magnitude.
- In mathematics, a function is defined on a vector space that maps each vector to its magnitude (a scalar quantity).
- In Machine learning, we use them to measure the distance between multiple data points in different dimensions.
- For example, the euclidean distance formula is the most often used formula to calculate the distance between any two points.
- Let x be any vector, then the norm of x is given by: $\|x\|$.

Properties of Vector Norm

- **Positive Definiteness:** The norm (or the magnitude) of a vector is always positive.
 - The norm value is zero if and only if the vector is zero.
 - i.e., $\|x\| \geq 0$, for all non-zero vector x .
- **Triangle Inequality:** The sum of the norm of two vectors is always less than the sum of their (individual) norm.
 - If x and y are two vectors, then,
 - $\|x + y\| \leq \|x\| + \|y\|$
- **Homogeneity:** Let k be any constant, and x be any vector then:
 - $\|kx\| = |k|\|x\|$



■ **Application of Norms of Vector in Machine Learning**

- Used to evaluate the error in the model, i.e., the distance between the observed value and predicted value.
- Vector norms are used in regularization, which is a technique to prevent overfitting in the model.

Types of vector norms

■ **L^p Norm**

- It is defined as the sum of the absolute value of the elements of the vector, raised to the pth power and then taking the pth root of the result.
- The L_p norm is written as $||x_p||$, where x is the vector and p is a positive real number.

$$L^p - Norm = ||x_p|| = \left(\sum_{i=1}^n |x_i|^p \right)^{\frac{1}{p}}, \text{ where } p > 0$$

Types of vector norms

■ L1-Norm

- If $p = 1$, then the L^p norm is known as L1 – Norm. It is defined as the sum of the absolute value of the components of a given vector.
- L1-Norm is also known as **Manhattan** or the **Taxicab** Norm. It is typically the distance a taxi will have to cover from the origin.
- It is used in LASSO Regression.
- Let $\mathbf{x}$ be any vector, then the L1 norm of a vector is given by:

$$||\mathbf{x}||_1 = \left(\sum_{i=1}^n |x_i|^1 \right)^{\frac{1}{1}} = \sum_{i=1}^n |x_i|$$

Types of vector norms

■ L2-Norm

- If $p = 2$, then the L^p norm is known as L2 – Norm. It is defined as the square root of the sum of the squared vector values.
- It is also known as Euclidean Norm.
- It is mainly used in KNN Algorithm It is mainly used in KNN Algorithm and Ridge Regression.
- Let $\mathbf{x}$ be any vector, then the L2 norm of a vector is given by:

$$\|\mathbf{x}\|_2 = \left(\sum_{i=1}^n |x_i|^2 \right)^{\frac{1}{2}}$$

Types of vector norms

- **L infinity-Norm**

- If $p = \text{infinity}$, then the L^p norm is known as the L infinity norm. It is defined as the absolute value of the largest component of the vector.
- It is also called the **Max Norm**

$$||x||_{\infty} = \max_i (|x_i|)$$

Inner Product

- The inner product (or dot product) of two vectors is an operation that performs the element-wise multiplication and sum of the elements of each vector.
- The inner product is a frequently used operation in pattern recognition and, by extension, machine learning.
- The inner product is represented by
$$\mathbf{x} \cdot \mathbf{y} = \mathbf{x}^T \mathbf{y} = x_1 y_1 + x_2 y_2 + x_3 y_3 + \dots$$
- where $(\cdot)^T$ represents a transpose operation.
- The transpose of a matrix involves interchanging rows with columns or vice versa.
- For example, the transpose of a column vector would be a row vector, thus altering the dimensions from $n \times 1$ to $1 \times n$.

Inner Product

- The inner product of two vectors produces a scalar and is often considered a measure of similarity between vectors.
- Example:

$$\mathbf{x}^T \mathbf{y} = [1 \quad -3 \quad 0 \quad 2 \quad 5] \begin{bmatrix} 0 \\ 1 \\ 4 \\ -4 \\ 0 \end{bmatrix} = 0 - 3 + 0 - 8 + 0 = -11$$

Vector length

- The magnitude of a vector is the length of the vector.
- The length of the vector AB is denoted as |AB|.
- the length of the vector $a = \{a_x ; a_y\}$ can be found using the following formula:

$$|\vec{a}| = \sqrt{a_x^2 + a_y^2}$$

- Example 1. Find the length of the vector $a = \{2; 4\}$. **Solution:**

- Example 2. Find the length of the vector $a = \{2; 4\}$. **Solution:**

$$|\vec{a}| = \sqrt{2^2 + 4^2} = \sqrt{4 + 16} = \sqrt{20} = 2\sqrt{5}.$$

$$|\vec{a}| = \sqrt{3^2 + (-4)^2} = \sqrt{9 + 16} = \sqrt{25} = 5.$$

The Distance Between Two Vectors

Distance between $\vec{u}$ and $\vec{v}$ is

$$d(\vec{u}, \vec{v}) = \|\vec{u} - \vec{v}\| = \sqrt{(u_1 - v_1)^2 + (u_2 - v_2)^2 \dots (u_n - v_n)^2}.$$

Example :

Determine the Euclidean distance between $\vec{u} = (2, 3, 4, 2)$ and $\vec{v} = (1, -2, 1, 3)$.

Applying the formula given above we get that:

$$d(\vec{u}, \vec{v}) = \|\vec{u} - \vec{v}\| = \sqrt{(2 - 1)^2 + (3 + 2)^2 + (4 - 1)^2 + (2 - 3)^2}$$

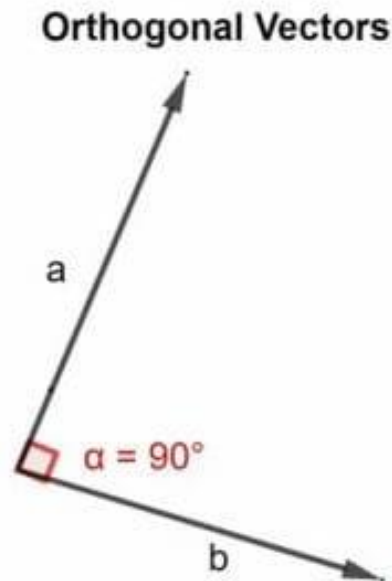
$$d(\vec{u}, \vec{v}) = \|\vec{u} - \vec{v}\| = \sqrt{1 + 25 + 9 + 1}$$

$$d(\vec{u}, \vec{v}) = \|\vec{u} - \vec{v}\| = \sqrt{36}$$

$$d(\vec{u}, \vec{v}) = \|\vec{u} - \vec{v}\| = 6$$

Orthogonal vectors

- vectors are called orthogonal if they are perpendicular to each other, and after performing the dot product analysis, the product they yield is zero.



Orthogonal vectors

■ *Example 1:*

Check whether the vectors $\mathbf{a} = \mathbf{i} + 2\mathbf{j}$ and $\mathbf{b} = 2\mathbf{i} - \mathbf{j}$ are orthogonal or not.

Solution:

For checking whether the 2 vectors are orthogonal or not, we will be calculating the dot product of these vectors:

$$\mathbf{a} \cdot \mathbf{b} = (1 \cdot 2) + (2 \cdot (-1))$$

$$\mathbf{a} \cdot \mathbf{b} = 2 - 2$$

$$\mathbf{a} \cdot \mathbf{b} = 0$$

Hence as the dot product is 0, so the two vectors are orthogonal.

Orthogonal vectors

■ *Example 2:*

Are the vectors $\mathbf{a} = (3, 2)$ and $\mathbf{b} = (7, -5)$ orthogonal?

Solution

For checking whether the 2 vectors are orthogonal or not, we will be calculating the dot product of these vectors:

$$\mathbf{a} \cdot \mathbf{b} = (3 \cdot 7) + (2 \cdot (-5))$$

$$\mathbf{a} \cdot \mathbf{b} = 21 - 10$$

$$\mathbf{a} \cdot \mathbf{b} = 11$$

Since the dot product of these 2 vectors is not a zero, these vectors are not orthogonal.

Symmetric Matrix

- A symmetric matrix is a square matrix that is equal to its transpose matrix.
- The transpose matrix of any given matrix A can be given as A^T .
- A symmetric matrix A , therefore, satisfies the condition, $A = A^T$.
- This can be represented as:

If $B = [b_{ij}]_{n \times n}$ is the symmetric matrix, then $b_{ij} = b_{ji}$ for all i and j or $1 \leq i \leq n$, and $1 \leq j \leq n$.

Here, n is any natural number.

b_{ij} is an element at position (i, j) which is i th row and j th column in matrix B and

b_{ji} is an element at position (j, i) which is j th row and i th column in matrix B .

Symmetric Matrix

- Symmetric Matrix Example:

$$B = \begin{bmatrix} 2 & 3 & 6 \\ 3 & 4 & 5 \\ 6 & 5 & 9 \end{bmatrix}$$

$$B^T = \begin{bmatrix} 2 & 3 & 6 \\ 3 & 4 & 5 \\ 6 & 5 & 9 \end{bmatrix}$$

- Here, we can see that, $B^T = B$. For example, $b_{12} = b_{21} = 3$, and $b_{13} = b_{31} = 6$. Thus, B is a symmetric matrix.
- **Positive Matrix:** A positive matrix is a real matrix. A positive matrix is a real or integer matrix for which each matrix element is a positive number.

Determinant of a Matrix

- Determinant of a Matrix is defined as the function that gives the unique output (real number) for every input value of the matrix.
- the scalar value computed for a given square matrix.
- Determinant of the matrix is considered the scaling factor that is used for the transformation of a matrix.
- Determinant of a matrix is useful for finding the solution of a system of linear equations, the inverse of the square matrix, and others.
- The determinant of only square matrices exists.

Determinant of a Matrix

■ **Symbol of Determinant**

- Let's take any square matrix A , then the determinant of A is denoted as **$\det A$** (or) **$|A|$** .
- Determinant is also denoted by the symbol **Δ** .

Properties of Determinants

- **Reflection Property:** Value of the determinant remains unchanged even after rows and columns are interchanged. That determinant of a matrix and its transpose remains the same.
- **Repetition Property/Proportionality Property:** If any two rows or any two columns of a determinant are identical, then the value of the determinant becomes zero.
- **Switching Property:** If any two rows or columns of a determinant are interchanged, then the sign of the determinant changes.

- **Example:**

$$\begin{vmatrix} 3 & 3 & 0 \\ 2 & 1 & 1 \\ 5 & 0 & 1 \end{vmatrix}$$

Solution:

$$\det. A = [3 \times \{(1 \times 1) - (0 \times 1)\}] - [3 \times \{(2 \times 1) - (5 \times 1)\}] + [0 \times \{(2 \times 0) - (5 \times 1)\}]$$

$$= \{3 \times (1 - 0)\} - \{3 \times (2 - 5)\} + 0$$

$$= [3 - \{3(-3)\} + 0]$$

$$= (3 + 9)$$

$$= 12$$

Now, Interchanging Row 1 with Row 2, determinant will be:

$$\begin{vmatrix} 2 & 1 & 1 \\ 3 & 3 & 0 \\ 5 & 0 & 1 \end{vmatrix}$$

$$\det. A = [2 \times \{(3 \times 1) - (0 \times 0)\}] - [1 \times \{(3 \times 1) - (5 \times 0)\}] + [1 \times \{(3 \times 0) - (5 \times 3)\}]$$

$$= (6 - 3 - 15)$$

$$= -12$$

- **Scalar Multiple Property:** If each element of a row (or a column) of a determinant is multiplied by a constant k , then its value gets multiplied by k

$$\begin{vmatrix} ka & kb \\ c & d \end{vmatrix} = k \begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

- **Sum Property:** If some or all elements of a row or column can be expressed as the sum of two or more terms, then the determinant can also be expressed as the sum of two or more determinants.

$$\begin{vmatrix} a_1 + \lambda_1 & a_2 + \lambda_2 & a_3 + \lambda_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} + \begin{vmatrix} \lambda_1 & \lambda_2 & \lambda_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

Determinant of a 1×1 Matrix

Let $X = [a]$ be the matrix of order one, then its determinant is given by $\det(X) = a$.

Determinant of a 2×2 Matrix

The determinant of any 2×2 square matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}_{2 \times 2}$

is calculated using the formula $|A| = ad - bc$.

Example: Find the Determinant of $A =$

Solution:

Determinant of $A = \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix}_{2 \times 2}$ is calculated as,

$$|A| = \begin{vmatrix} 3 & 2 \\ 2 & 3 \end{vmatrix}$$

$$\begin{aligned} |A| &= 3 \times 3 - 2 \times 2 \\ &= 9 - 4 \\ &= 5 \end{aligned}$$

■ Determinant of a 3×3 Matrix

- The determinant of any 3×3 square matrix A

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

- is calculated using the formula,

$$|A| = a(ei - fh) - b(di - fg) + c(dh - eg)$$

- Example : Evaluate

$$\begin{vmatrix} 102 & 18 & 36 \\ 1 & 3 & 4 \\ 17 & 3 & 6 \end{vmatrix}$$

Trace of a matrix

- Trace of a matrix is defined as the sum of the principal diagonal elements of a square matrix.
- It is usually represented as $\text{tr}(A)$, where A is any square matrix of order “ $n \times n$.”
- Let us consider a square matrix of order “ 3×3 ,” as shown in the figure given below, $a_{11}, a_{12}, a_{13}, \dots, a_{32}, a_{33}$ are the entries of the given matrix A .
- Now, the trace of matrix “ A ” is equal to the sum of its principal diagonal elements, i.e., a_{11}, a_{22} , and a_{33} .

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$
$$\text{tr}(A) = a_{11} + a_{22} + a_{33}$$

Trace of a matrix

- If A is a square matrix of order “ $n \times n$,” then the trace of matrix A is equal to the sum of the main diagonal elements.

$$\text{tr}(A) = a_{11} + a_{22} + a_{33} + \dots + a_{nn}$$

Properties of Trace of a Matrix

Let us consider two square matrices A and B of the same order.

- The sum of the traces of the matrix A and the matrix B is equal to the trace of the matrix that is obtained by the sum of the matrices A and B.

$$\text{tr}(\mathbf{A}) + \text{tr}(\mathbf{B}) = \text{tr}(\mathbf{A} + \mathbf{B})$$

- The trace of a given matrix and its transpose are the same.

$$\text{tr}(\mathbf{A}) = \text{tr}(\mathbf{A}^T)$$

- If A is a matrix of order “ $m \times n$ ” and B is a matrix of order “ $n \times m$,” then the trace of AB is equal to the trace of BA.

$$\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$$

- The trace of an identity matrix of order “ $n \times n$ ” is n.

$$\text{tr}(\mathbf{I}_n) = n$$

- The trace of a zero or null matrix of any order is zero.

$$\text{tr}(\mathbf{O}) = 0$$

Example 1: Calculate the trace of the matrix given below.

$$\begin{bmatrix} 1 & -3 & 4 & 7 \\ 2 & 11 & -9 & 6 \\ 17 & 8 & -5 & 3 \\ 5 & -22 & 14 & -4 \end{bmatrix}$$

From the given matrix,

$$b_{11} = 1, b_{22} = 11, b_{33} = -5, \text{ and } b_{44} = -4.$$

We know that,

$$\begin{aligned} \text{tr}(A) &= b_{11} + b_{22} + b_{33} + b_{44} \\ &= 1 + 11 + (-5) + (-4) \\ &= 12 - 5 - 4 = 12 - 9 = 3 \end{aligned}$$

Thus, the trace of the given matrix B is 3.

Example 2: Calculate the trace of the matrix given below.

$$A = \begin{bmatrix} 0 & 11 & 19 & 41 & 23 \\ 9 & 24 & -6 & 17 & 14 \\ -8 & 13 & 7 & 25 & 0 \\ -19 & 31 & 15 & -5 & 23 \\ 5 & 10 & -4 & 32 & 16 \end{bmatrix}$$

From the given matrix,

$$a_{11} = 0, a_{22} = 24, a_{33} = 7, a_{44} = -5, \text{ and } a_{55} = 16.$$

We know that,

$$\begin{aligned} \text{tr}(A) &= a_{11} + a_{22} + a_{33} + a_{44} + a_{55} \\ &= 0 + 24 + 7 + (-5) + 16 \\ &= 47 - 5 = 42 \end{aligned}$$

Thus, the trace of the given matrix A is 42.

Eigenvalues and Eigenvectors

- Eigenvalues and Eigenvectors are the scalar and vector quantities associated with Matrix used for linear transformation.
- The vector that does not change even after applying transformations is called the Eigenvector and the scalar value attached to Eigenvectors is called Eigenvalues.
- Eigenvectors are the vectors that are associated with a set of linear equations.
- The equation for eigenvalue is given by

$$\mathbf{Av} = \lambda \mathbf{v}$$

Where, A is the matrix

v is associated eigenvector

λ is scalar eigenvalue

Eigenvalues and Eigenvectors

■ **Steps for calculating Eigenvalues**

Step 1. Form the matrix $B = (A - \lambda I)$

Step 2. Create an equation using: Determinant of $B = 0$.
It is a polynomial equation in λ , $p(\lambda) = 0$

Step 3. Solve $p(\lambda) = 0$. These are the eigenvalues of the matrix A

■ **Steps for Calculating Eigenvectors**

We can easily solve the original equation $Ax = \lambda x$ for eigenvectors using the eigenvalue.

Step 1. Find eigenvalues λ of A

Step 2. For **each** λ , form homogeneous system of linear equations $(A - I\lambda)x = 0$.

Step 3. Solve the above equations to get eigenvectors for λ

■ Example1:

Find eigenvalues of the 2 x 2 matrix $A = \begin{pmatrix} 1 & 2 \\ 3 & 0 \end{pmatrix}$

Solution: We use the following equation to calculate eigenvalues:

Determinant of $(A - \lambda I) = 0$

Step 1. Form Matrix

Step 2. Create Equation

$$A - \lambda I = \begin{pmatrix} 1 - \lambda & 2 \\ 3 & 0 - \lambda \end{pmatrix}$$

$$\det(A - \lambda I) = 0$$

$$(1 - \lambda)(-\lambda) - 6 = 0$$

$$\lambda^2 - \lambda - 6 = 0$$

Step 3. Solve

$$(\lambda - 3)(\lambda + 2) = 0$$

$$\lambda = 3 \text{ or } \lambda = -2$$

Eigenvalues are 3 and -2

■ Example 2

Find eigenvectors of $A = \begin{pmatrix} 1 & 2 \\ 3 & 0 \end{pmatrix}$.

Step 1. Find Eigenvalues

We have already calculated the eigenvalues as -2 and 3.

First we find the eigenvector for $\lambda = -2$.

Step 2. Form Equations for $\lambda = -2$

$$(A - I\lambda) =$$

$$\begin{pmatrix} 1 - (-2) & 2 \\ 3 & 0 - (-2) \end{pmatrix} = \begin{pmatrix} 3 & 2 \\ 3 & 2 \end{pmatrix}$$

Step 3. Solve Equations

We now solve $(A - I\lambda)x = 0$

$$\begin{pmatrix} 3 & 2 \\ 3 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0$$

We get the equation:

$$3x_1 + 2x_2 = 0$$

$$x_1 = \frac{-2}{3}x_2$$

We can choose any value for x_2 to calculate the eigenvector.

For $x_2 = 3$ we get $x_1 = -2$

Eigenvector is $(-2, 3)^T$

Any value of x_2 results in an eigenvector. They are all scaled by a factor. **Any scalar multiple of an eigenvector is also an eigenvector.**

Eigenvector for $\lambda = 3$:

Again we calculate $(A - I\lambda) =$

$$\begin{pmatrix} 1 - 3 & 2 \\ 3 & 0 - 3 \end{pmatrix} = \begin{pmatrix} -2 & 2 \\ 3 & -3 \end{pmatrix}$$

We now solve $(A - I\lambda)x = 0$

$$\begin{pmatrix} -2 & 2 \\ 3 & -3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0$$

We get the equation:

$$-2x_1 + 2x_2 = 0$$

$$x_1 = x_2$$

For $x_2 = 1$ we get $x_1 = 1$

Eigenvector is $(1, 1)^T$

- **Example: Find the eigenvalues and the eigenvector for the matrix $A =$**

$$\begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}$$

Solution:

If eigenvalues are represented using λ and the eigenvector is represented as $v = \begin{bmatrix} a \\ b \end{bmatrix}$

Then the eigenvector is calculated by using the equation,
 $|A - \lambda I| = 0$

$$\begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1- & 2 \\ 5 & 4- \end{bmatrix}$$


$$(1-\lambda)(4-\lambda) - 2.5 = 0$$

$$4 - \lambda - 4\lambda + \lambda^2 - 10 = 0$$

$$\lambda^2 - 5\lambda - 6 = 0$$

$$\lambda^2 - 6\lambda + \lambda - 6 = 0$$

$$\lambda(\lambda-6) + 1(\lambda-6) = 0$$

$$(\lambda-6)(\lambda+1) = 0$$

$$\lambda = 6 \text{ and } \lambda = -1$$

Thus, the eigenvalues are 6, and -1.

Then the respective eigenvectors are,

For $\lambda = 6$

$$(A - \lambda I)v = 0$$

$$\begin{bmatrix} 1 - 6 & 2 \\ 5 & 4 - 6 \end{bmatrix} \cdot \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -5 & 2 \\ 5 & -2 \end{bmatrix} \cdot \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$-5a - 2b = 0$$

$$5a + 2b = 0$$

simplifying the above equation we get,

$$5a = -2b$$

The required eigenvector is,

$$\begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$$

For $\lambda = -1$

$$(A - \lambda I)v = 0$$

$$\begin{bmatrix} 1 - (-1) & 2 \\ 5 & 4 - (-1) \end{bmatrix} \cdot \begin{bmatrix} a \\ b \end{bmatrix} = 0$$

$$\begin{bmatrix} 2 & 2 \\ 5 & 5 \end{bmatrix} \cdot \begin{bmatrix} a \\ b \end{bmatrix}$$

$$a + 2b = 0$$

$$5a + 5b = 0$$

simplifying the above equation we get,

$$a = -b \quad \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

The required eigenvector is,

Then the eigenvectors of the given 2×2 matrix are

$$\begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}, \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$